Homework 1: Range compression of FMCW radar

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Instructions

- The outputs of the homework are a set of MATLAB files and a PowerPoint presentation, which will be the start of a full oral exam. The PowerPoint presentation must be limited to a **maximum of 10 slides**.
- You can work in groups to share ideas, doubts, and possible solutions. However, the final presentation is **individual**.
- The text doesn't ask you to plot specific figures or show specific equations. It is up to the student to decide which figure will better convey the information to the reader and to produce it.
- For each homework, there are lessons dedicated to classwork. During these hours, you can work and ask the instructor questions. No questions are allowed outside these lessons.
- The text defines requirements and parameters. Missing information - if any - are a free choice of the engineer.
- The deadline: **29/11/2026** (included).
- Compress all files (MATLAB and presentation) in a ZIP package and call it `name_surname_HW1_XXXXXXXX.zip`, where XXXXXXXX is the ID number of the student (e.g. `marco_manzoni_HW1_10455829.zip`)
- The ZIP must be sent by email at marco.manzoni@polimi.it and in Carbon copy (Cc) to andrea.montiguarnieri@polimi.it
- The oral discussion of the homework will be scheduled using Doodle, where you can reserve a time slot, typically between January and the end of February. At the oral, you will present and discuss your homework assignments.

1 Geometry and signal model

1.1 Geometry

The geometry under analysis is the one depicted in Figure 1. A Frequency Modulated Continuous Wave (FMCW) radar (the blue dot) is pointed with the antenna towards the y direction (the maximum of the directivity is towards y). Two targets are present in the scene, represented by the orange and green dots. Target 1 (orange dot) has coordinates $(x, y) = (-10m, 100m)$, while Target 2 (green dot) has coordinates $(x, y) = (30m, 50m)$. Assume a 2D scene in which targets lie at an elevation angle $\theta = 0$. The radar parameters are provided in Table 1.

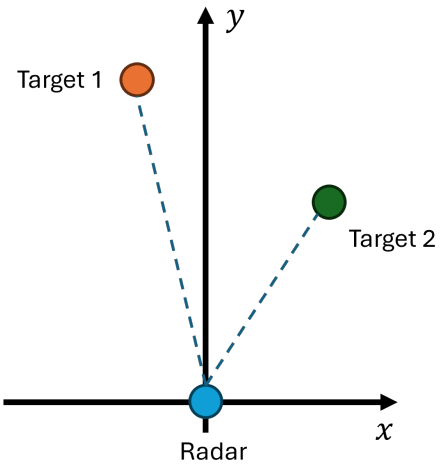


Figure 1: System geometry.

Table 1: Parameters of the system.

Parameter	Value	Unit
Radar Cross Section (σ)	5, 3	m^2
Carrier frequency (f_0)	77	GHz
Azimuth beam-width $\Delta\psi$ (Angle to first null)	60	deg
Elevation beam-width $\Delta\theta$ (Angle to first null)	60	deg
Antenna pattern	$\text{sinc}^2(\psi/\Delta\psi)\text{sinc}^2(\theta/\Delta\theta)$	-
Bandwidth (B)	1	GHz
Pulse length (T_p)	55	μ s
Transmitted power (P_{TX})	10	dBm

1.2 FMCW signal model

Assume a stationary radar transmits a chirp signal toward a stationary target in the scene. The transmitted signal with bandwidth B and duration T_p can be modeled as:

$$s_{TX}(t) = \sqrt{P_{TX}}g(t)\exp\left(j2\pi f_0 t\right) \tag{1}$$

in which $g(t) = \exp\left(j\pi\alpha t^2\right)\text{rect}(t/T_p)$ is the chirp function with chirp rate $\alpha = B/T_p$, f_0 is the carrier frequency and P_{TX} the transmitted power.

The received signal from a stationary target at distance R_0 can be written as:

$$s_{RX}(t) = \sqrt{P_{RX}}g(t-t_0)\exp\left(j2\pi f_0(t-t_0)\right) \tag{2}$$

where $t_0 = 2R_0/c$ is two-way travel time, also called *delay*.

In a Frequency Modulated Continuous Wave (FMCW) radar, the received signal is multiplied by the complex conjugate of a unit-amplitude replica of the transmitted chirp, leading to the demodulated and deramped signal:

$$\begin{aligned} s_D(t) = &\left[\sqrt{P_{RX}}\,\text{rect}\left(\frac{t-t_0}{T_p}\right)\exp\left(j\pi\alpha(t-t_0)^2\right)\exp\left(j2\pi f_0(t-t_0)\right)\right] \\ &\cdot\left[\text{rect}\left(\frac{t}{T_p}\right)\exp\left(-j\pi\alpha t^2\right)\exp\left(-j2\pi f_0 t\right)\right] \end{aligned} \tag{3}$$

$$\approx \sqrt{P_{RX}}\,\text{rect}\left(\frac{t}{T_p}\right)\exp\left(-j2\pi\alpha t_0 t\right)\exp\left(j\pi\alpha t_0^2\right)\exp\left(-j2\pi f_0 t_0\right) \tag{4}$$

This signal is then processed to estimate the target range R_0. This processing is often called *range-compression*.

1.3 Received signal power

The incident power density at the target is:

$$S_i = \frac{P_{TX}G_{TX}f_{TX}(\theta_0,\psi_0)}{4\pi R_0^2} \quad [W/m^2] \tag{5}$$

in which G_{TX} is the gain of the transmitting antenna, and $f_{TX}(\theta_0,\psi_0)$ is the directivity function of the antenna computed at the angular position of the target (θ_0,ψ_0). The antenna gain can be estimated from the angular extent of the main lobe using a suitable approximation. The equivalent isotropic scattered power associated with the backscattered field is:

$$P_s = S_i\sigma \quad [W] \tag{6}$$

in which σ is the so-called radar cross section (RCS) of the target. The incident power density at the antenna of the receiver is:

$$S_r = \frac{P_s}{4\pi R_0^2} \quad [W/m^2] \tag{7}$$

To obtain the received power, we have to integrate the power density over the equivalent area of the receiving antenna and multiply by its directivity function:

$$P_{RX} = S_r A_e f_{RX}(\theta_0, \psi_0) \quad [W] \tag{8}$$

Assuming identical transmit and receive antennas, $G_{RX} = G_{TX}$ and $f_{RX} = f_{TX}$. The effective receiving area at boresight is $A_e = \lambda^2 G_{RX}/(4\pi)$.

2 Questions

1. Explain how the target ranges can be estimated from Eq. (4). Identify the approximation used to derive this equation, state the conditions under which it is valid, and discuss the consequences when these conditions are not met. Determine the minimum sampling frequency required to sample the beat signal without aliasing, specifying your sampling assumptions. Compare this requirement with that of a pulsed radar whose received signal is sampled after demodulation, without deramping.

2. Implement a MATLAB simulation of the specified scenario. Define the target positions and compute their ranges, look angles, and received powers. Generate the demodulated and deramped signal, then perform range compression. Verify that the peaks occur at the expected target ranges.

3. Assume that each target moves with a constant velocity. Derive the corresponding signal model for $s_D(t)$ and explain how target motion affects the range-compressed response. Implement a MATLAB simulation with different target velocities and illustrate the observed effects. *Hint:* Consider at least two distinct effects. For each effect, define a quantitative criterion for treating it as negligible and derive the corresponding conditions on target velocity.

4. *Optional:* Propose and implement a method to improve range estimation for moving targets. Validate your method using the simulated scenario and discuss its assumptions and limitations.